

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

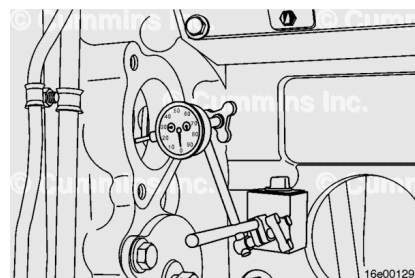
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the starter air to prevent accidental engine starting. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/56/56-012-022.html>)
- Remove the transmission, clutch, and all related components. See equipment manufacturer service information.

Deflection of the flexplate **must** be measured when the generator is decoupled in order to check the flexplate suitability for reuse.

Before removing the generator from the engine, position a dial test indicator with peak retaining indication to measure maximum displacement of the flexplate at the moment of generator decoupling.

Record the measured value.

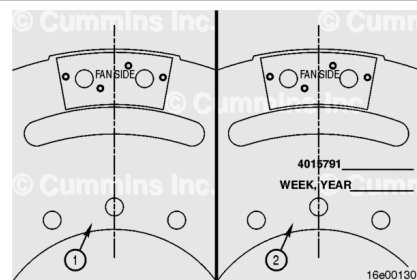
If deflection measurement can **not** be taken due to lack of suitable dial test indicator or access restrictions in vehicle, replace the flexplate with new one.



Displacement limits for reuse of the flexplate are:

- Previous generation flexplates (1) - 5.6 mm (0.22 inch).
- New generation flexplates (2) - 10.0 mm (0.39 inch).
- Insulated flexplates do **not** have displacement limits.

Note : On new generation flexplates (2), part number and production week/year are etched centrally under the stamped or etched marking indicating “FAN SIDE” located close to outer edge of the flexplate. Correctly installed flexplates will have the “FAN SIDE” marking,



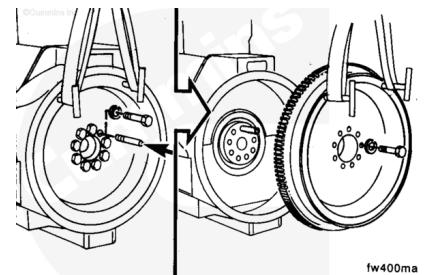
facing outward from the engine block toward the generator fan.

If the flexplate deflection is greater than the limit, the flexplate **must** be replaced.

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The use of a guide pin will aid the disassembly procedure.

Remove the capscrews and the flexplate.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not disassemble the flexplate assembly. Improper assembly of the flexplate can cause engine damage. Refer to Procedure 016-999 in Section F.

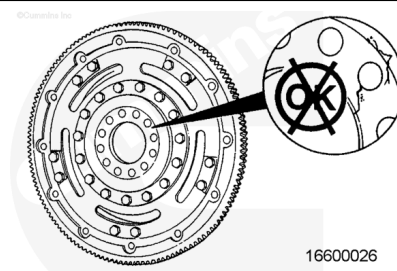
Clean the flexplate with solvent and dry with compressed air.

Check the ring gear for damaged or worn teeth. If the ring gear is damaged or worn, the assembly **must** be replaced.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.

⚠ WARNING ⚠

Cummins Inc. does not recommend resurfacing of the flexplate. Resurfacing of the flexplate can cause possible personal injury, property damage, or cause it to be too thin and result in breakage.



Use the dye penetrant method to check the area of the mounting holes for cracks. There **must not** be any cracks in the flexplate mounting area.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.

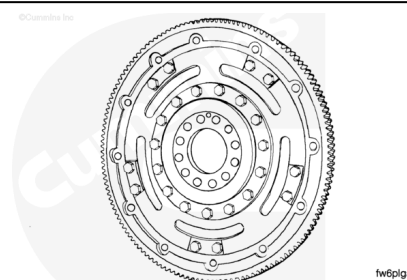
Check the tightness of the twenty-five capscrews.

Torque Value:

Flexplate capscrews 290 n·m [214 ft-lb]

If replacement is required, use **only** the correct grade 10.9 capscrews released by Cummins Inc.

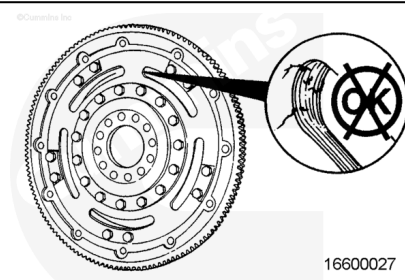
If the capscrews require replacement, remove and replace them one at a time. Do **not** disassemble the flexplate.



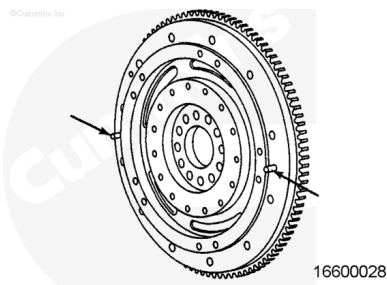
The flexplate can consist of as many as nine separate steel plates.

Check closely for cracks.

If any of the steel plates are cracked or otherwise damaged, the engine flexplate **must** be replaced.



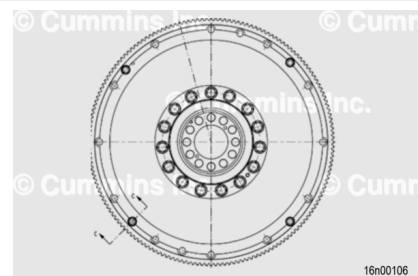
Check the two tapered dowels for damage. If the dowels are bent or broken, they **must** be replaced.



For 60L engines fitted with an insulated flexplate

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

Do not disassemble the flexplate assembly. Improper assembly of the flexplate can cause engine damage. Refer to Procedure 016-999 (</qs3/pubsys2/xml/en/procedures/56/56-016-999.html>) in Section F.

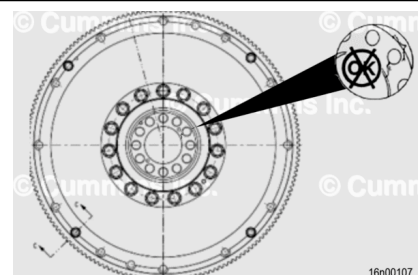
Clean the flexplate with solvent and dry with compressed air.

Check the ring gear for damaged or worn teeth. If the ring gear is damaged or worn, the assembly **must** be replaced.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.

⚠ WARNING ⚠

Cummins Inc. does not recommend resurfacing of the flexplate. Resurfacing of the flexplate can cause possible personal injury, property damage, or cause it to be too thin resulting in breakage.



Use the dye penetrant method to check the area of the mounting holes for cracks. There **must not** be any cracks in the flexplate mounting area.

Check the mounting pilot for damage. If the pilot is damaged, the flexplate **must** be replaced.

Check the tightness of the 19 capscrews.

Torque Value:

4 x outer flexplate capscrews 257 n•m [190 ft-lb]

Torque Value:

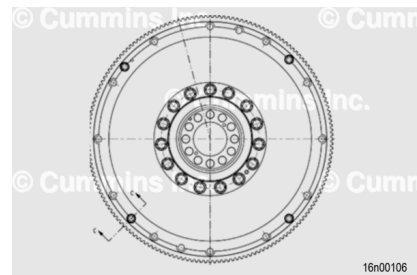
15 x inner flexplate capscrews 293 n•m [216 ft-lb]

If replacement is required, use **only** the correct grade 10.9 capscrews released by Cummins Inc.

If the capscrews require replacement, remove and replace them one at a time. Do **not** disassemble the flexplate.

Mechanical damage can be caused to the flexplate because of impact from heavy subjects like tools, or due to overloading. If any of the following are present, replace the flexplate immediately:

- Visible fiber failures or delamination (layer separation).
- Audible cracking during slow rotation of the part.
- Rotational movement between the crankshaft mating flange and external ring. Observe in resin area for movement.

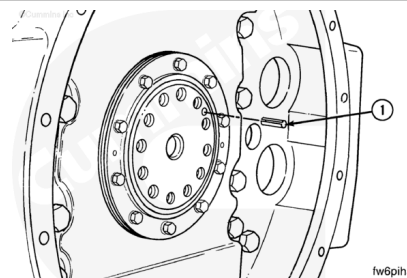


Install

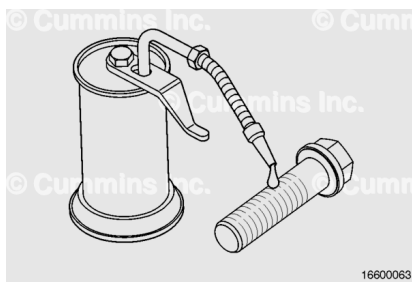
Note : The flexplate is **not** the same for the QSK45 and QSK60 engines. The valve and injector adjustment marks that are stamped on the parts are different.

Install the roll pin (1) in the crankshaft.

The pin is required to align the timing marks that are on the flexplate.

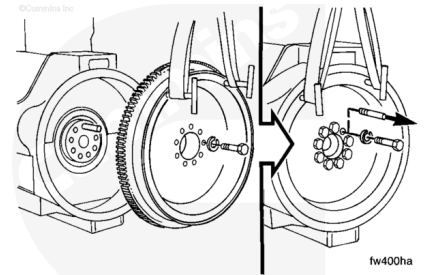


Lubricate the flexplate mounting capscrew threads and flange head surfaces with clean engine oil.



⚠ WARNING ⚠

The flexplate mounting cap screws are manufactured to a precise specification with rolled threads. If replacement is required, only the parts released by Cummins® can be used, otherwise a malfunction may result which could cause personal injury.



The use of a guide bolt will aid the assembly procedure.

Check to be sure the mating surfaces of the crankshaft and flexplate are clean, dry, and free of nicks or burrs.

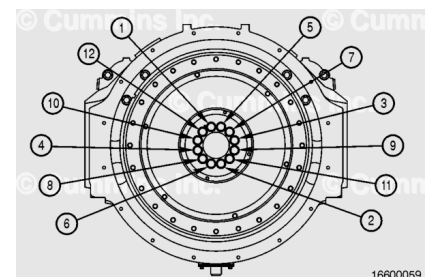
Align the hole in the flexplate for the roll pin with the roll pin in the crankshaft.

Install the flexplate and the cap screws. The flexplate **must** fit firmly against the crankshaft.

Use the following steps to tighten the cap screws in the sequence shown.

Torque Value:

1. 200 n•m [148 ft-lb]
2. 380 n•m [280 ft-lb]
3. 685 n•m [505 ft-lb]



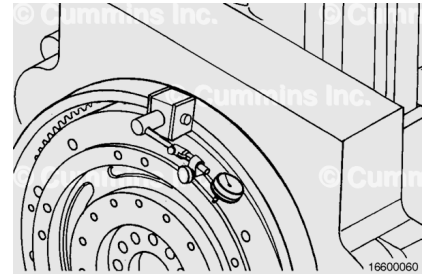
Measure

Note : The crankshaft **must** be pushed or pulled in the same direction each time a measurement is made in order to obtain an accurate reading.

Bar the engine from the front of the crankshaft for both radial and face measurements. The barring device will cause the flexplate to distort if barred from the rear, and will also cause the crankshaft to walk up the main bearing sidewall.

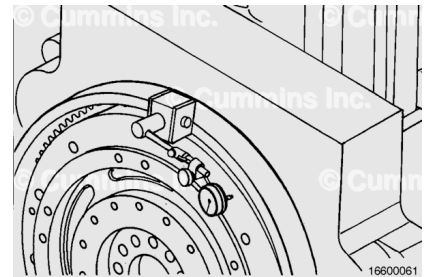
Radial Runout

- Attach an indicator to the flywheel housing with the indicator tip on the flexplate as illustrated. Check the indicator while rotating the engine.
 - Cummins® Society of Automotive Engineers (SAE) "0" flexplate: Maximum Radial Runout is 0.08 mm [0.003 in]
 - Cummins® SAE "00" flexplate: Maximum Radial Runout is 0.13 mm [0.005 in]
 - Cummins® insulated flexplate: Maximum Radial Runout is 0.25 mm [0.010 in]
- If the runout is **not** within specification, the pilot on the flexplate is **not** seated correctly on the crankshaft. If the pilot is damaged, the entire flexplate assembly **must** be replaced.



Face Runout

- Attach an indicator to the flywheel housing with the indicator tip on the face of the flexplate. Check the indicator while rotating the engine.
 - Cummins® SAE "0" flexplate: Maximum Face Runout is 0.89 mm [0.035 in]
 - Cummins® SAE "00" flexplate: Maximum Face Runout is 1.00 mm [0.040 in]
 - Cummins® insulated flexplate: Maximum Face Runout is 1.5 mm [0.060 in]
- If the alignment is **not** within specification, check for foreign material between the flexplate and the crankshaft.
- Two 3/8 x 16 inch threaded holes are provided in the adapter to aid in the removal of the assembly.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the transmission, clutch, and all related components. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Connect the starter air. Refer to Procedure 012-022 in Section 12.
(/qs3/pubsys2/xml/en/procedures/56/56-012-022.html)
- Start the engine. Check equipment operation.